IBM APPLIED DATA SCIENCE CAPSTONE PROJECT

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APRIL 6, 2026

OUTLINE

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- Methodology
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Photo credit: https://www.spacelaunchschedule.com/wp-content/uploads/2026/02/20240214_Starlink_7-14_prelaunch_small-678x452-9.jpg

EXECUTIVE SUMMARY

SUMMARY OF METHODOLOGIES

- Data collection and wrangling
- Data analysis
- Data visualization
- Machine learning prediction

SUMMARY OF RESULTS

- Exploratory data analysis results
- Interactive analysis demo in screenshots
- Predictive analysis results

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INTRODUCTION

SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars while other providers cost upward of 165 million dollars. Much of the savings is because SpaceX can reuse the first stage. Therefore if we can determine if the first stage will land, we can determine the cost of a launch. Based on public information and machine learning models, we are going to predict if SpaceX will reuse the first stage.

Questions to be answered

- How do variables such as payload mass, launch site, number of flights, and orbits affect the success of the first stage landing?
- Does the rate of successful landings increase over time?
- What is the best algorithm to be used for binary classification in this case?

METHODOLOGY

Methodologies

1. Data collecting and wrangling

   Collecting: SpaceX API, Web scraping

   Wrangling: Apply filters, address missing values, use one-hot encoding to encode categorical features

2. Exploratory data analysis (EDA)

   Pandas and NumPy

   SQL

   Matplotlib and Seaborn

   Folium

   Dash

3. Machine learning prediction

   Logistic regression

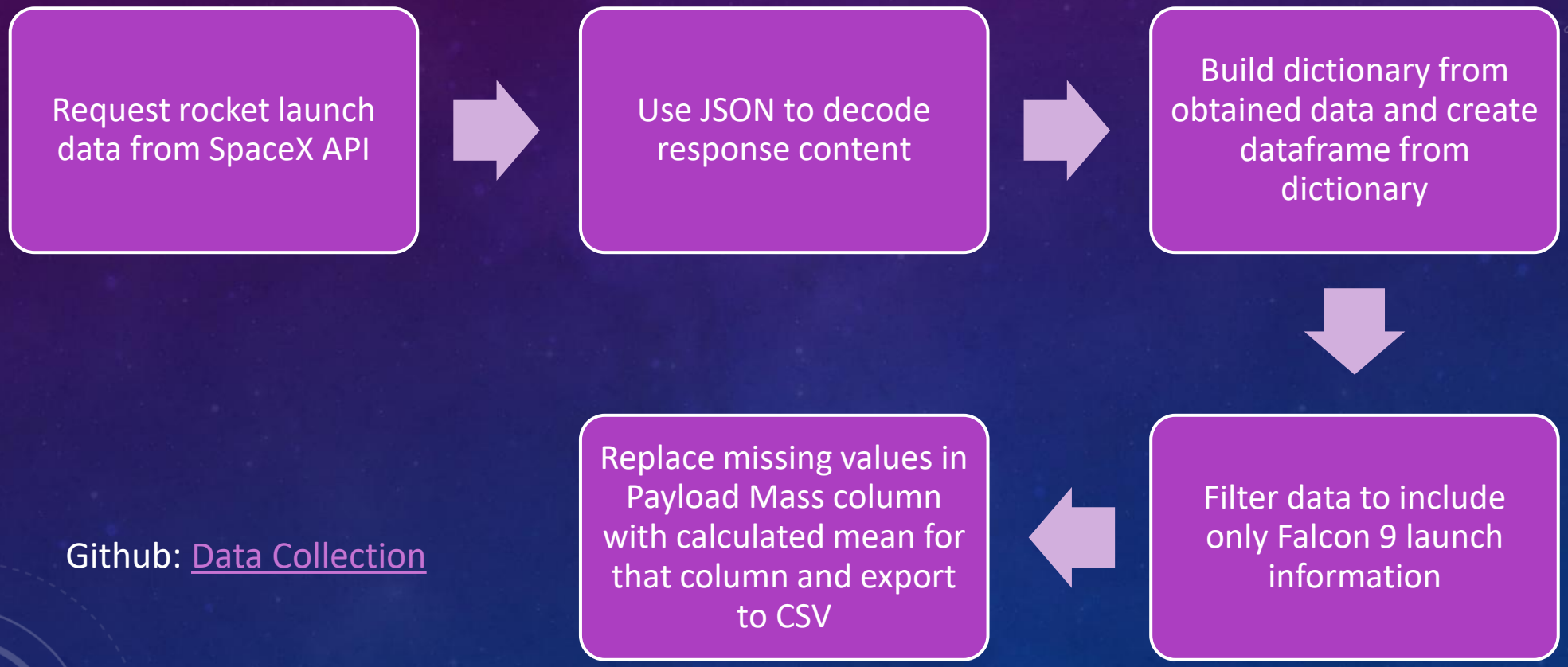
   Support vector machine (SVM)

   Decision tree

   K-nearest neighbors (KNN)

METHODOLOGY 1
DATA COLLECTING

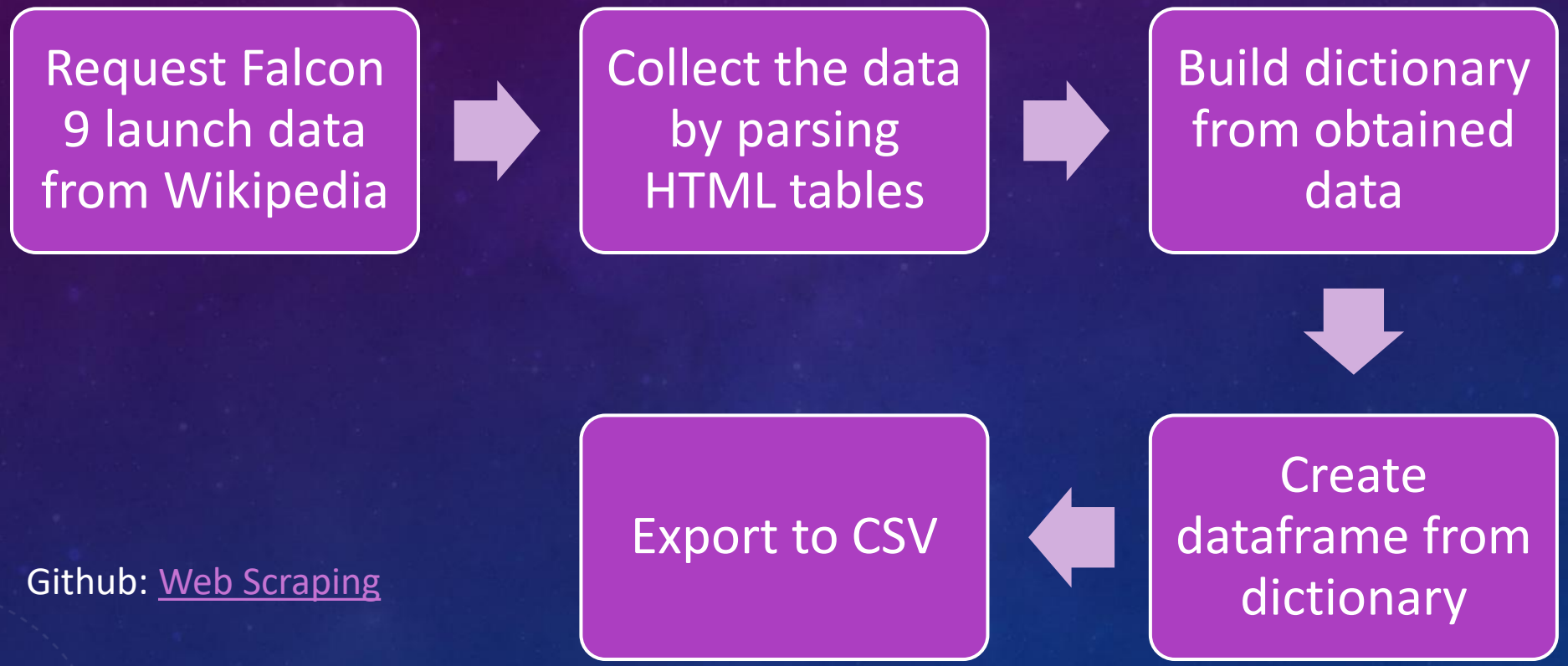
FLOWCHART OF STEPS TAKEN IN DATA COLLECTION USING API



Github: Data Collection

METHODOLOGY 1
DATA COLLECTING

FLOWCHART OF STEPS TAKEN IN DATA COLLECTION USING WEB SCRAPING



Github: Web Scraping

METHODOLOGY 1 WRANGLING

WRANGLING AND FORMATTING

Perform exploratory data analysis

Calculate number of launches per site

Calculate number and occurrence of each orbit

Calculate number and occurrence of mission outcome per orbit type

Add column 'Class' classifying 0 if a launch is failed and 1 if it is successful

Github: Data Wrangling

METHODOLOGY 2
EXPLORATORY DATA ANALYSIS (EDA)

PYTHON LIBRARIES

- ### PANDAS AND NUMPY

  Functions from the pandas and NumPy libraries are used to plot charts to determine relationships between variables:

  The number of launches on each launch site

  The number and occurrence of each orbit

  The number and occurrence of each mission outcome

- ### MATPLOTLIB AND SEABORN

  Functions from the Matplotlib and Seaborn libraries are used to visualize the findings with scatterplots, bar charts, and line charts.

  The plots and charts are used to understand more about the relationships between several features, such as:

  The relationship between flight number and launch site

  The relationship between payload mass and launch site

  The relationship between success rate and orbit type

Github: EDA Visualization

METHODOLOGY 2
EXPLORATORY DATA ANALYSIS (EDA)

PYTHON LIBRARIES

- FOLIUM

  Functions from the Folium libraries are used to visualize the data through interactive maps, including:

  Mark all launch sites on a map

  Successful and failed launches for each site

  Distances between a launch site and locations as the nearest city, railway, or highway

  Github: Site Location Folium

- DASH

  Functions from Dash are used to generate an interactive site with input toggled through a dropdown menu and a range slider

  Using a pie chart and a scatterplot, the interactive site shows:

  The total success launches from each launch site

  The correlation between payload mass and mission outcome (success or failure) for each launch site

  Github: Dash

METHODOLOGY 2
EXPLORATORY DATA ANALYSIS (EDA)

SQL

The data is queried using SQL to answer several questions including:

- The names of the unique launch sites in the space mission
- Displaying 5 records of Launch Site names that begin with CCA
- The total payload mass carried by boosters launched by NASA (CRS)
- The average payload mass carried by booster version F9 v1.1
- Total number of successful and failed mission outcomes
- Names of the booster versions which have carried the maximum payload mass
- Date of first successful landing outcome in ground pad
- Names of boosters which have success in drone landings with payload mass between 4000-6000
- Count of landing outcomes between the dates 2010-06-04 and 2017-03-20
- Failed landing outcomes in drone ship with their booster versions and launch site names in 2015

Github: SQL

METHODOLOGY 3
PREDICTIVE ANALYSIS

FUNCTIONS FROM THE SCIKIT-LEARN LIBRARY ARE USED TO CREATE OUR MACHINE LEARNING MODELS.

THE MACHINE LEARNING PREDICTION PHASE INCLUDES THE FOLLOWING STEPS:

- Standardizing the data
- Splitting the data into training and test data
- Creating machine learning models, which include:
  - Logistic regression
  - Support vector machine (SVM)
  - Decision tree
  - K nearest neighbors (KNN)
- Fit the models on the training set
- Find the best combination of hyperparameters for each model
- Evaluate the models based on their accuracy scores and confusion matrix

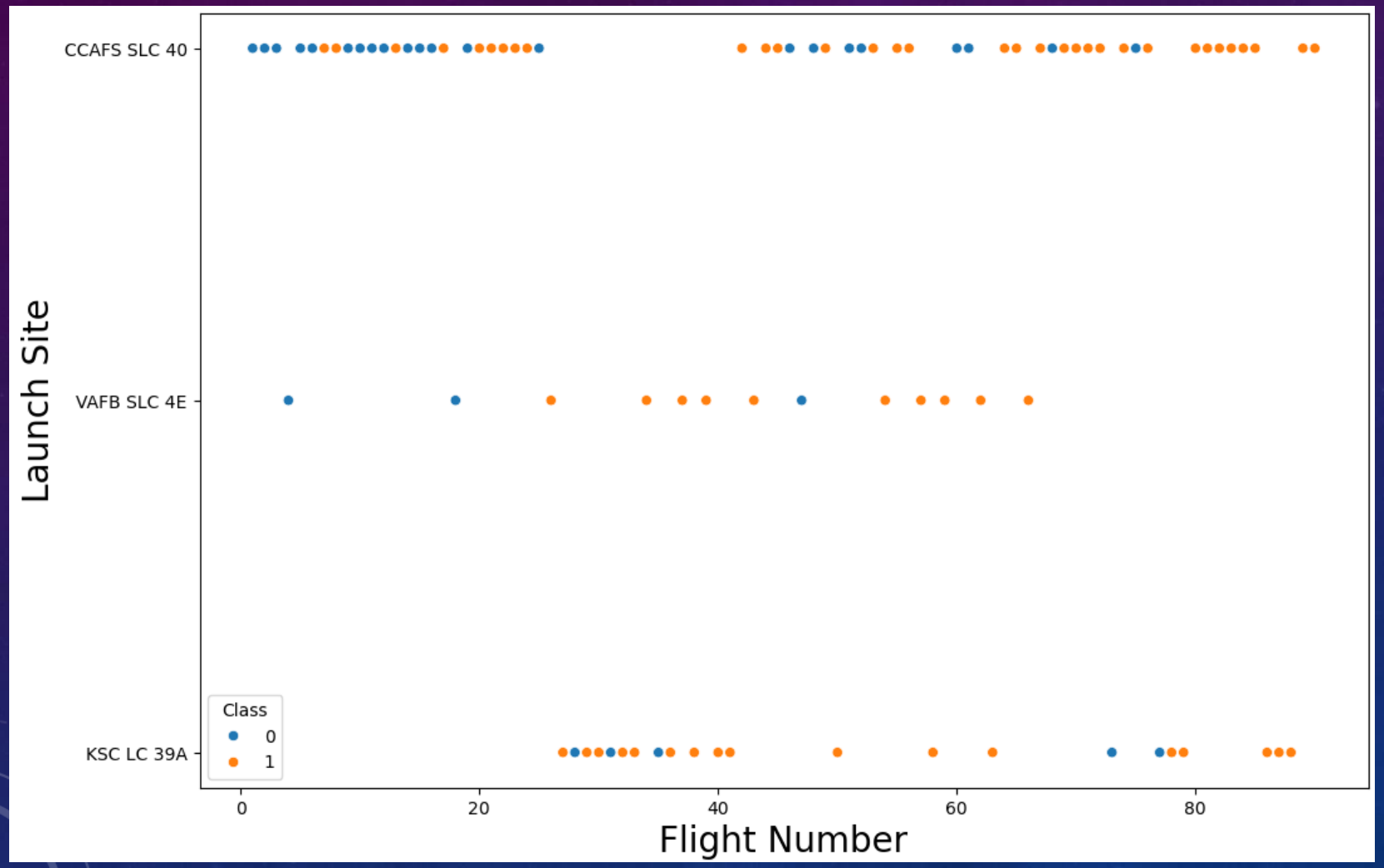
GITHUB: MACHINE LEARNING PREDICTIONS

RESULTS

Exploratory data analysis results
Interactive analytics demo in screenshots

EDA WITH VISUALIZATION

FLIGHT NUMBER VS LAUNCH SITE (CLASS 0 – FAILED, CLASS 1 – SUCCESSFUL)



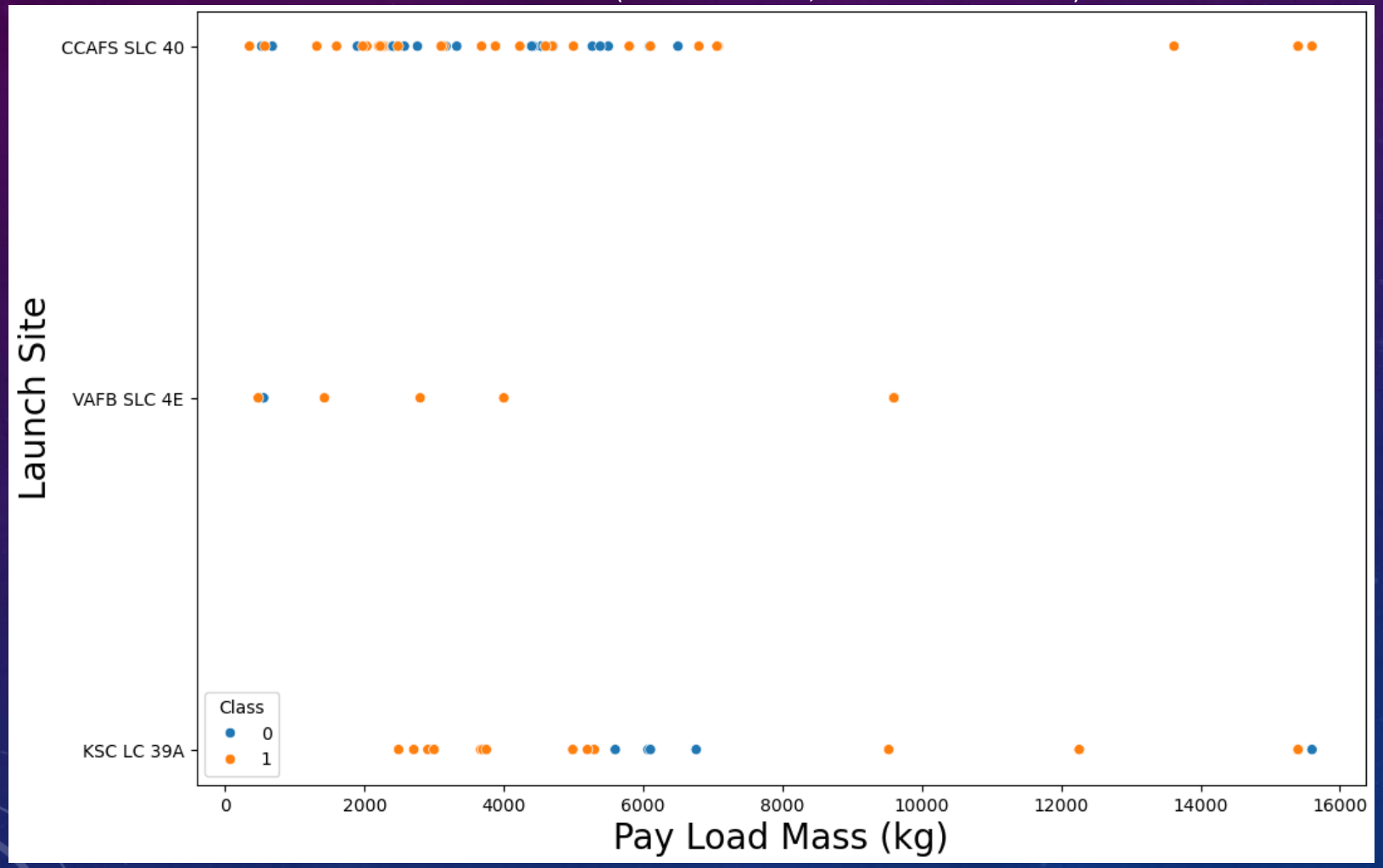
Explanation

The CCAFS SLC 40 launch site has about a half of all launches, but the VAFB SLC 4E and KSC LC 39A launch sites appear to have higher success rates.

Earliest flights failed while latest flights succeeded, leading to the conclusion that each new launch has a higher rate of success.

EDA WITH VISUALIZATION

PAYLOAD MASS VS LAUNCH SITE (CLASS 0 – FAILED, CLASS 1 – SUCCESSFUL)



Explanation

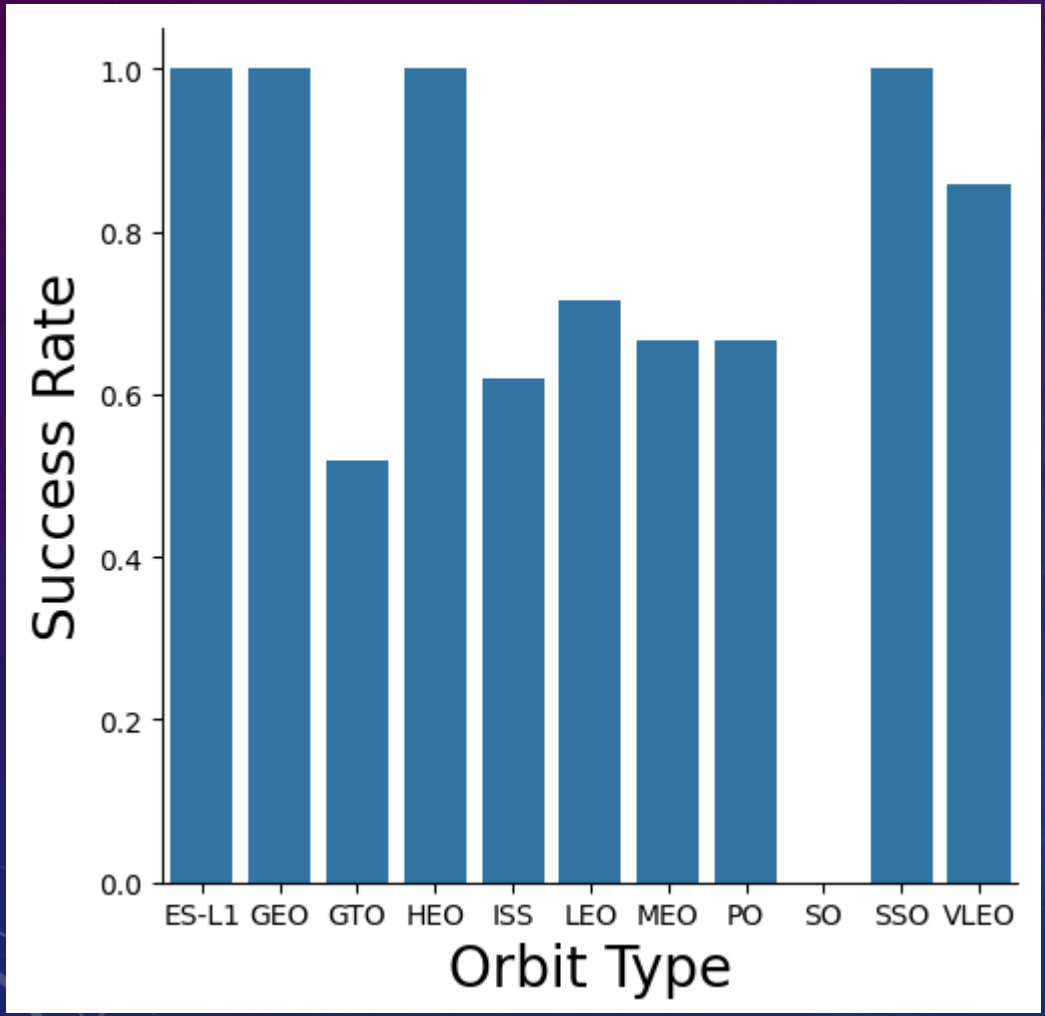
Launch site KSC LC 39A has a 100% success rate for payload mass under 5500 kg.

Most of the launches with a payload mass over 7000 kg were successful.

At each launch site the higher the payload mass the higher the success rate.

EDA WITH VISUALIZATION

SUCCESS RATE VS ORBIT TYPE



Explanation

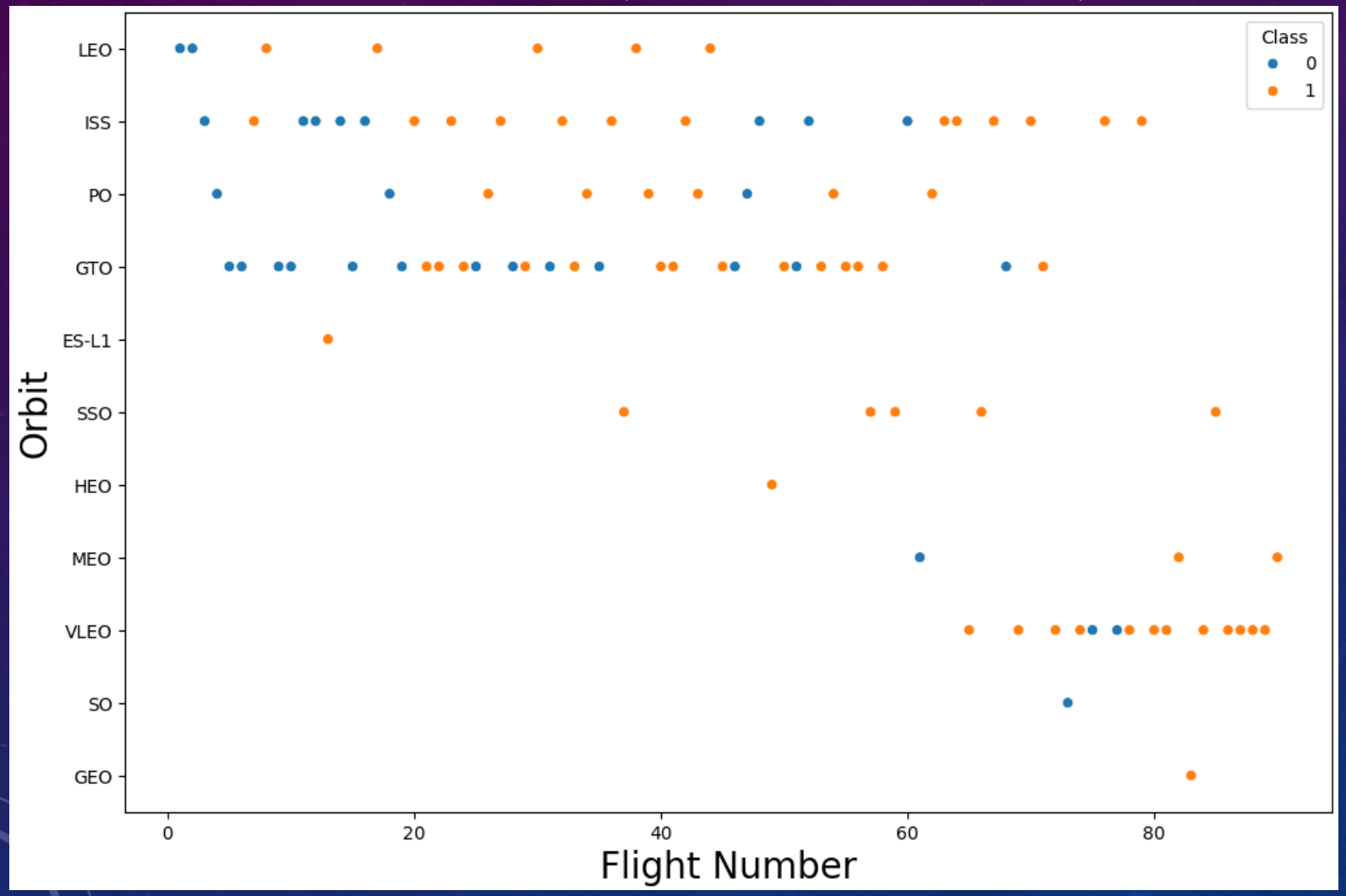
Orbits with 100% success rate
ES-L1
GEO
HEO
SSO

Orbits with success rate between 50% and 85%
GTO
ISS
LEO
MEO
PO

Orbits with 0% success rate
SO

EDA WITH VISUALIZATION

FLIGHT NUMBER VS ORBIT TYPE (CLASS 0 – FAILED, CLASS 1 – SUCCESSFUL)



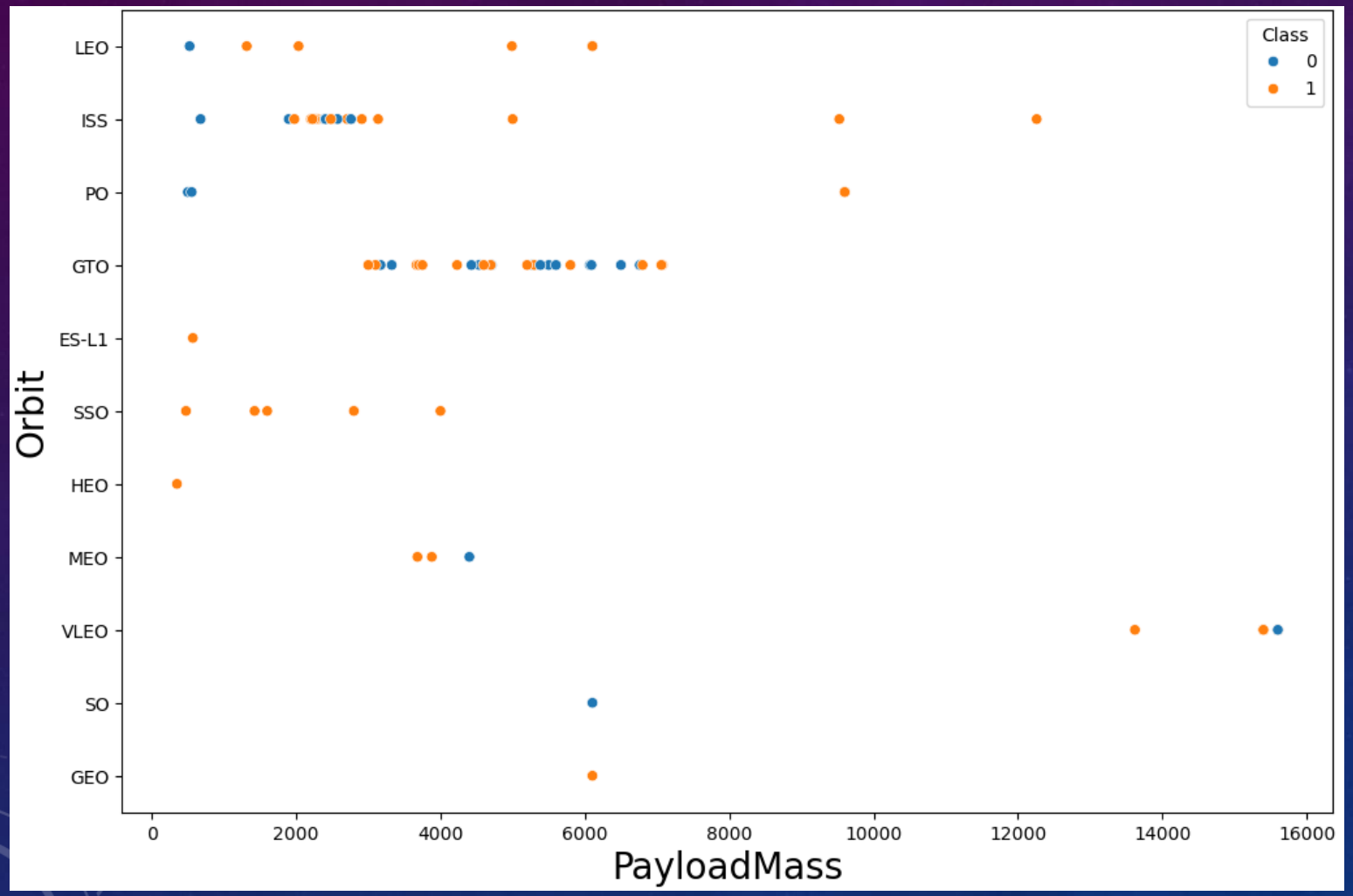
Explanation

In the LEO Orbit success appears to be related to the number of flights.

In the GTO orbit there does not appear to be a relationship between success and the number of flights.

EDA WITH VISUALIZATION

PAYLOAD MASS VS ORBIT TYPE (CLASS 0 – FAILED, CLASS 1 – SUCCESSFUL)



Explanation

Heavy loads appear to be more favorable for Polar, LEO, and ISS orbits.

For GTO orbit it's difficult to distinguish between successful and failed landings as both outcomes are present.

EDA WITH SQL

DISPLAYING THE UNIQUE LAUNCH SITES USING AN SQL QUERY

```
%%sql
SELECT
    DISTINCT launch_site
FROM
    SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

EDA WITH SQL

DISPLAYING 5 RECORDS OF LAUNCH SITE NAMES THAT BEGIN WITH CCA USING AN SQL QUERY

```
%%sql
SELECT
    *
FROM
    SPACEXTABLE
WHERE
    launch_site LIKE 'CCA%'
LIMIT
    5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

EDA WITH SQL

DISPLAYING TOTAL PAYLOAD MASS CARRIED BY BOOSTERS FROM NASA USING AN SQL QUERY

```
%%sql
SELECT
    SUM(payload_mass__kg_) AS total_payload_mass
FROM
   SPACEXTABLE
WHERE
    customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

total_payload_mass
45596

EDA WITH SQL

DISPLAYING AVERAGE PAYLOAD MASS CARRIED BY BOOSTER VERSION F9 V1.1 FROM NASA USING AN SQL QUERY

```
%%sql
SELECT
    AVG(payload_mass__kg_) AS average_payload_mass
FROM
    SPACEXTABLE
WHERE
    booster_version LIKE '%F9 v1.1%';
```

```
 * sqlite:///my_data1.db
Done.
```

average_payload_mass
2534.6666666666665

EDA WITH SQL

DISPLAYING FIRST SUCCESSFUL GROUND LANDING DATE USING AN SQL QUERY

```
%%sql
SELECT
    MIN(date) AS first_successful_landing
FROM
    SPACEXTABLE
WHERE
    landing_outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

first_successful_landing
2015-12-22

EDA WITH SQL

DISPLAYING SUCCESSFUL DRONE SHIP LANDING WITH PAYLOAD BETWEEN 4000 AND 6000 USING AN SQL QUERY

```
%%sql
SELECT
    booster_version
FROM
    SPACEXTABLE
WHERE
    landing_outcome = 'Success (drone ship)'
    AND payload_mass__kg_ BETWEEN 4000 AND 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

EDA WITH SQL

DISPLAYING TOTAL NUMBER OF SUCCESSFUL AND FAILURE MISSION OUTCOMES USING AN SQL QUERY

```
%%sql
SELECT
    mission_outcome,
    COUNT(*) AS total_number
FROM
    SPACEXTABLE
GROUP BY
    mission_outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	total_number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

EDA WITH SQL

DISPLAYING NAMES OF THE BOOSTER WHICH HAVE CARRIED THE MAXIMUM PAYLOAD MASS USING AN SQL QUERY

```
%%sql
SELECT
    booster_version
FROM
    SPACEXTABLE
WHERE
    payload_mass__kg_ = (
        SELECT
            MAX(payload_mass__kg_)
        FROM
            SPACEXTABLE
    );
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

EDA WITH SQL

DISPLAYING FAILED LANDING OUTCOMES IN DRONE SHIP WITH THEIR BOOSTER VERSIONS AND LAUNCH SITE NAMES IN 2015 USING AN SQL QUERY

```
%%sql
SELECT
    strftime('%m', date) AS month,
    date,
    booster_version,
    launch_site,
    landing_outcome
FROM SPACEXTABLE
WHERE
    landing_outcome = 'Failure (drone ship)'
    AND strftime('%Y', date) = '2015';
```

```
 * sqlite:///my_data1.db
Done.
```

month	Date	Booster_Version	Launch_Site	Landing_Outcome
01	2015-01-10	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	2015-04-14	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

EDA WITH SQL

DISPLAYING THE COUNT OF LANDING OUTCOMES BETWEEN THE DATES 2010-06-04 AND 2017-03-20 IN DESCENDING ORDER USING AN SQL QUERY

```
%%sql
SELECT
    landing_outcome,
    COUNT(*) AS count_outcomes
FROM
    SPACEXTABLE
WHERE
    date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    landing_outcome
ORDER BY
    count_outcomes DESC;
```

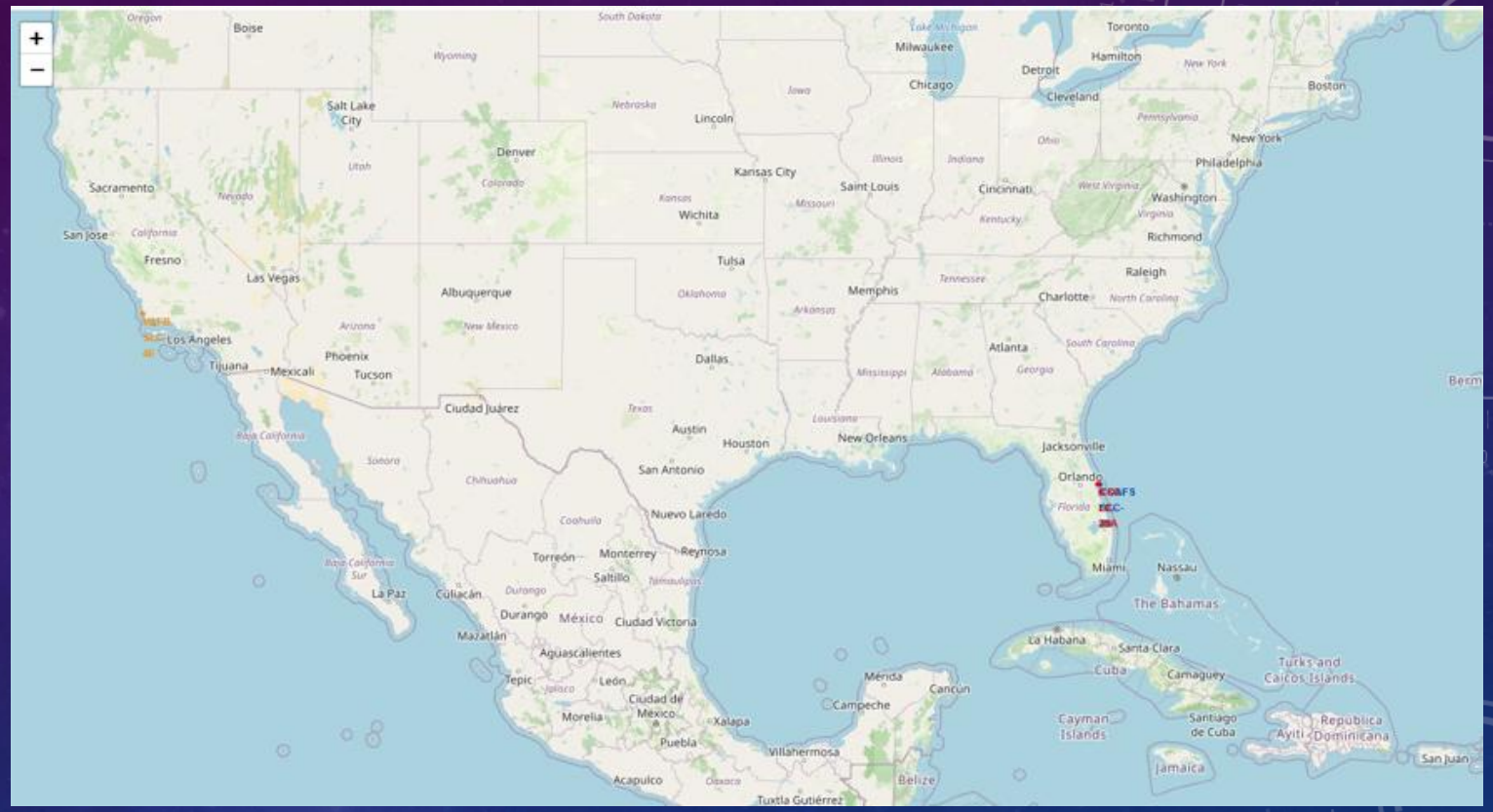
```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	count_outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

ALL LAUNCH LOCATION MARKERS ON GLOBAL MAP WITH FOLIUM

MOST OF THE LAUNCH SITES ARE IN PROXIMITY TO THE EQUATOR TO TAKE ADVANTAGE OF THE INCREASED ROTATIONAL VELOCITY AND LOW-INCLINATION AND GEOSTATIONARY ORBITS.

THEY ARE ALSO MORE CONCENTRATED ALONG COASTLINES TO MINIMIZE RISK AND SIMPLIFY RECOVERY OF HARDWARE. PROXIMITY TO THE OCEAN LETS RECOVERED BOOSTERS RETURN TO NEARBY DRONE SHIPS OR LANDING ZONES QUICKLY.



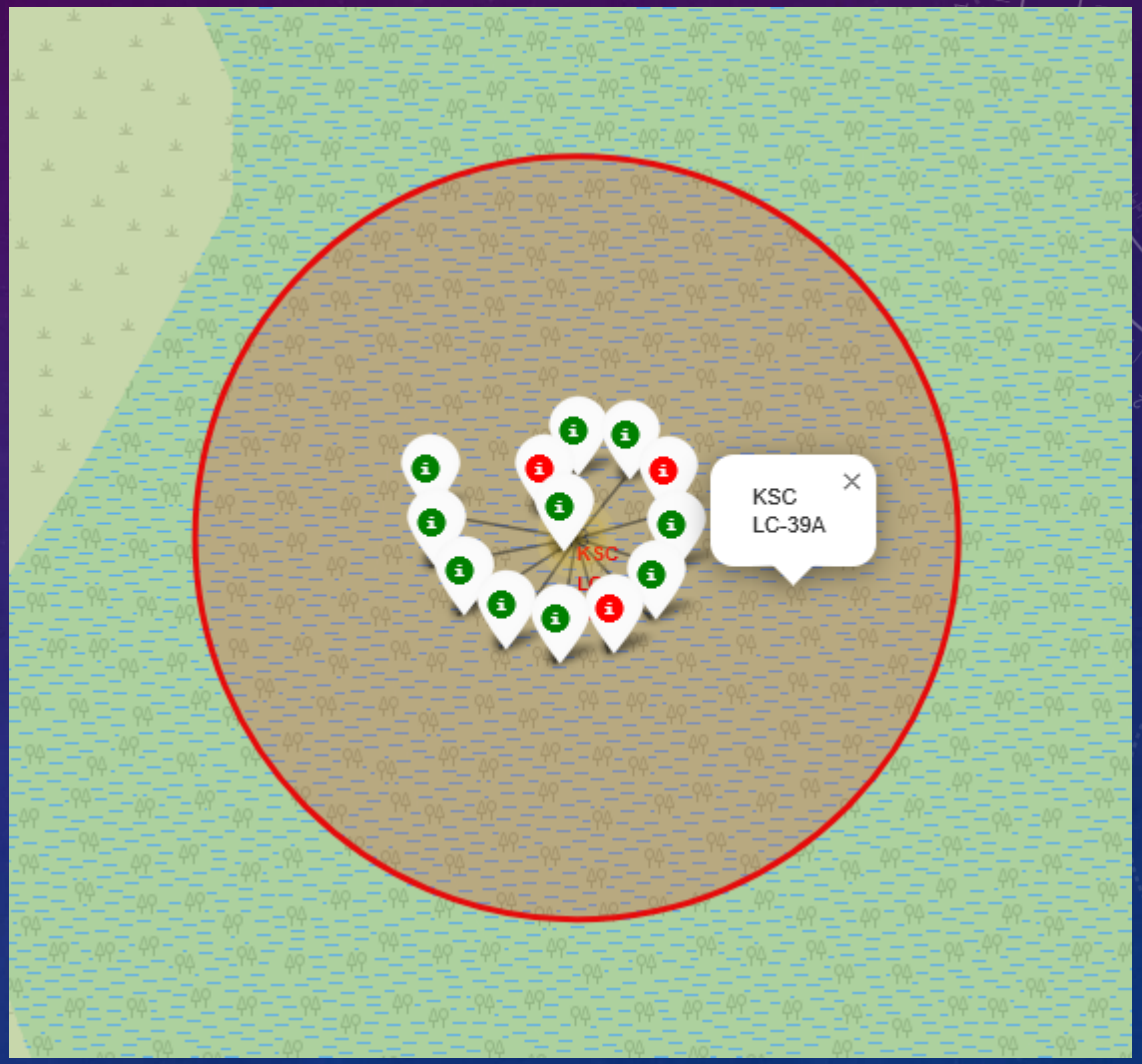
COLOR LABELED LAUNCH RECORDS ON MAP WITH FOLIUM

USING COLOR LABELS TO MARK LAUNCHES ALLOWS US TO EASILY IDENTIFY WHICH LAUNCH SITES HAVE RELATIVELY HIGH SUCCESS RATES.

GREEN MARKERS INDICATE SUCCESSFUL LAUNCHES

RED MARKERS INDICATE FAILED LAUNCHES

AS SHOWN IN THE IMAGE, LAUNCH SITE KSC LC-39A HAS A HIGH SUCCESS RATE.



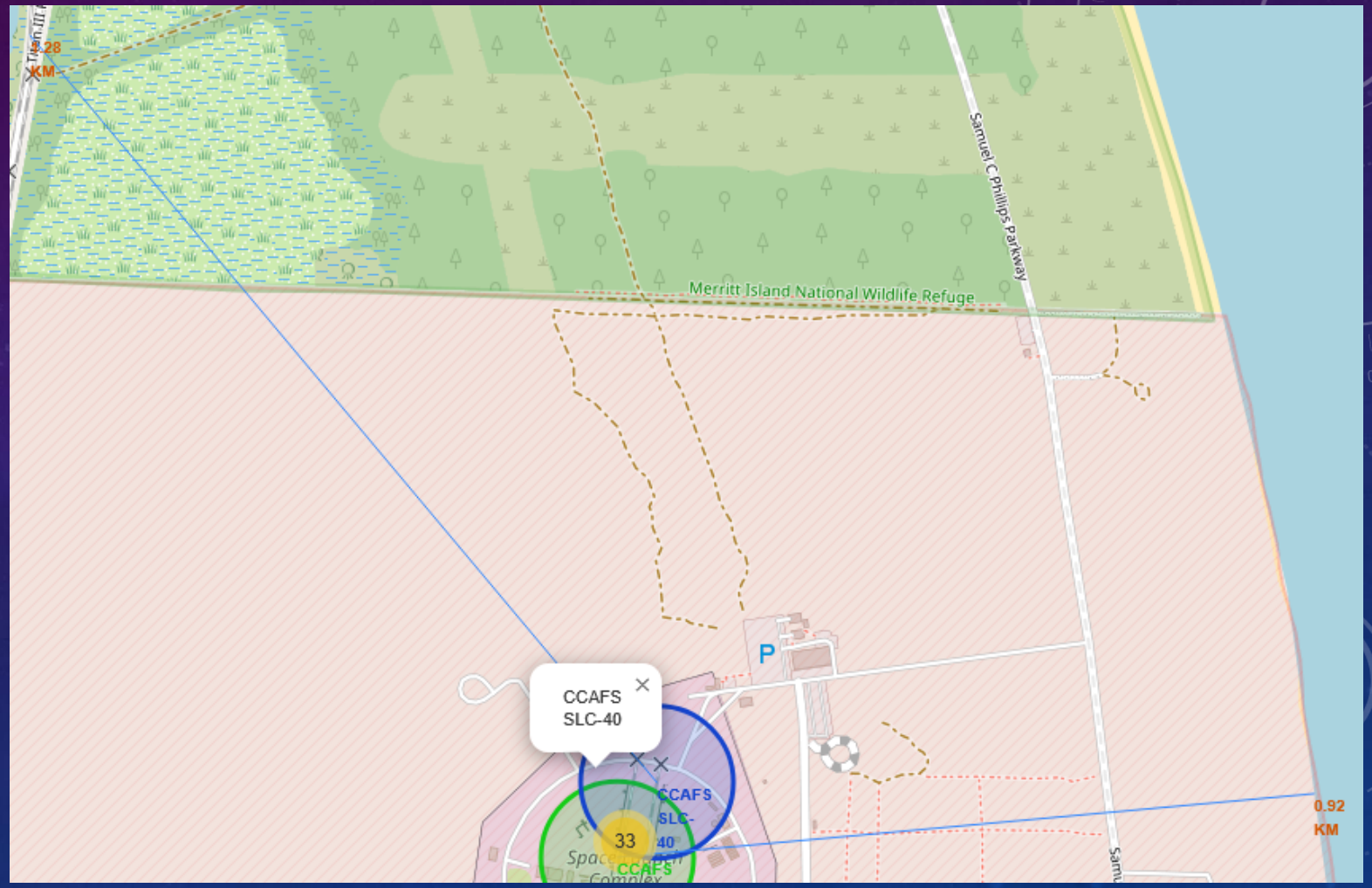
DISTANCE FROM LAUNCH SITE CCAFS SLC-40 TO PROXIMITIES ON MAP WITH FOLIUM

FROM THE VISUAL ANALYSIS OF THE LAUNCH SITE CCAFS SLC-40 WE CAN DETERMINE THE FOLLOWING:

CLOSEST RAILWAY: 1.28 KM

CLOSEST HIGHWAY: 0.83 KM

CLOSEST CITY: 13.06 KM



DASHBOARD WITH PLOTLY DASH

THE DASHBOARD IS COMPRISED OF TWO PLOTS

PIE CHART SHOWING LAUNCH SUCCESSES AND FAILURES

- INTERACTIVE DROPDOWN MENU ENABLES TOGGLING THROUGH ALL SITES
- ALLOWS QUICK VISUAL ASSESSMENTS AND COMPARISONS OF LANDING OUTCOMES BY LAUNCH SITE

SCATTER PLOT SHOWING PAYLOAD MASS VS. LANDING OUTCOME USING COLORS TO DISTINGUISH BOOSTER VERSIONS

- DROPDOWN MENU OFFERS THE SAME INTERACTIVITY AS THE PIE CHART
- ADDITIONAL SLIDER ALLOWS FILTERING BY PAYLOAD RANGE DATA

PREDICTIVE ANALYSIS
(CLASSIFICATION)

PREDICTIVE ANALYSIS USING MACHINE LEARNING

FOUR MACHINE LEARNING CLASSIFICATIONS MODELS WERE CONSTRUCTED USING THIS PROCESS TO PREDICT LANDING OUTCOME

- LOGISTIC REGRESSION
- SUPPORT VECTOR MACHINE
- DECISION TREE
- K-NEAREST NEIGHBOR

Standardize

Standardize the data using

```
transform = preprocessing.StandardScaler()
X = transform.fit(X).transform(X)
```

Split

Split the data into training and test data using

```
X_train, X_test, Y_train, Y_test =
train_test_split( X, Y, test_size=0.2,
random_state=2)
```

Create and Optimize

Create model object and parameters dictionary using

```
parameters ={"C":[0.01,0.1,1],'penalty':['l2'],
'solver':['lbfgs']}# l1 lasso l2 ridge
lr=LogisticRegression()
logreg_cv = GridSearchCV(lr, parameters, cv=10)
```

Train

Train the model using

```
logreg_cv.fit(X_train, Y_train)
```

Evaluate

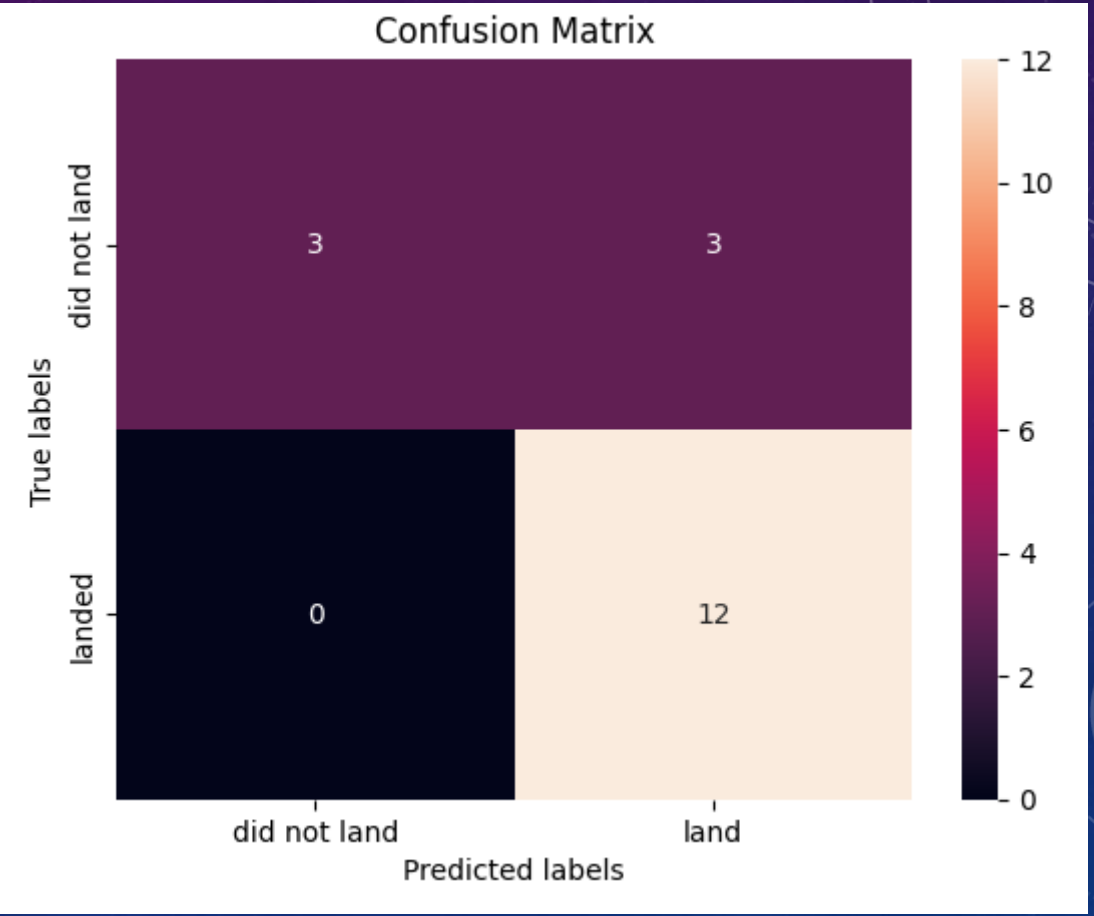
Evaluate the accuracy of the model using

```
lr_optimized = LogisticRegression(C=0.1, penalty='l2',
solver='lbfgs').fit(X_train, Y_train)
lr_score = lr_optimized.score(X_test, Y_test)
print('Accuracy Score: {}' .format(lr_score))
```

CONFUSION MATRIX

ALL FOUR MODELS HAD THE SAME THE CONFUSION MATRIX AND SIMILAR ACCURACY SCORES

18 OUTCOMES WERE PREDICTED AND EACH HAD 3 FALSE POSITIVES.



CONCLUSION

LANDING OUTCOME OF THE FIRST STAGE ROCKET CAN BE PREDICTED WITH 83.34% ACCURACY

VARIABLES AFFECTING LANDING OUTCOME

- LAUNCH SITE: GEOGRAPHY PLAYS AN IMPORTANT ROLE. LAUNCH SITES SHOULD BE AS CLOSE TO THE EQUATOR AND A COASTLINE AS POSSIBLE
- LAUNCH NUMBER: MORE LAUNCHES LEAD TO MORE EXPERIENCE AND HIGHER SUCCESS RATES
- PAYLOAD: HEAVIER PAYLOADS TEND TO HAVE MORE FAILED LAUNCHES

APPENDIX

GITHUB LINK TO MAIN REPOSITORY

HTTPS://GITHUB.COM/PWHEELIS/IBM-APPLIED-DATA-SCIENCE-CAPSTONE/TREE/MAIN